Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**6**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

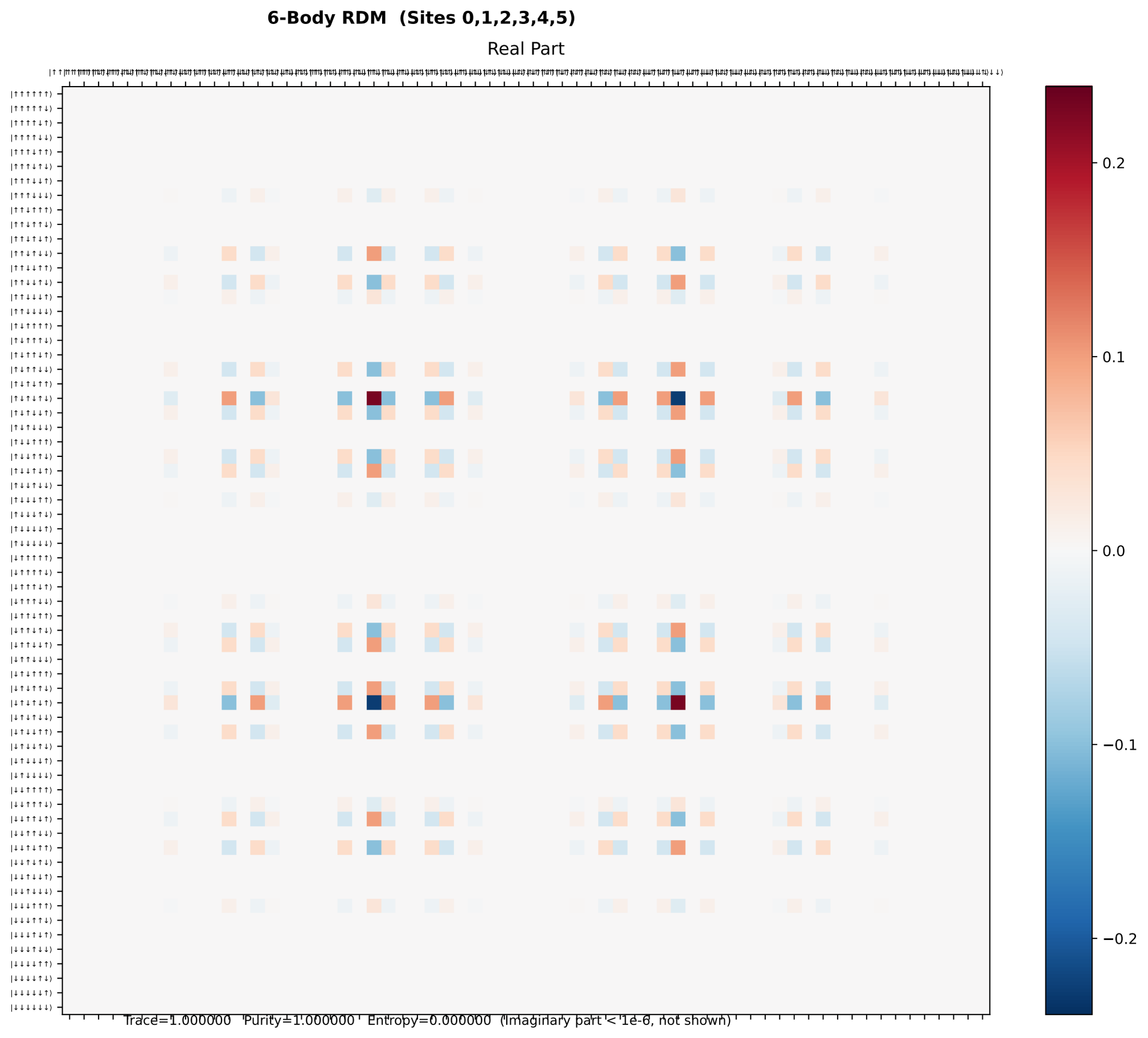
Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (6-body clusters)

Hilbert space dim : 2^6 = 64
Number of clusters : 6
Pure clusters : 6
Mixed clusters : 0
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = 0.958232 [min 0.958232 , max 0.958232]


6-Body RDM (Sites 0,1,2,3,4,5)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 0,1,2,3,4,5)

RDM Metrics
==========

```
  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -2.399988e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

Convex Roof Analysis
==========

```
  Lower Bounds:
    C_LB    = 0.45569925
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.95823220   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.95823220

  Status: nONE
```

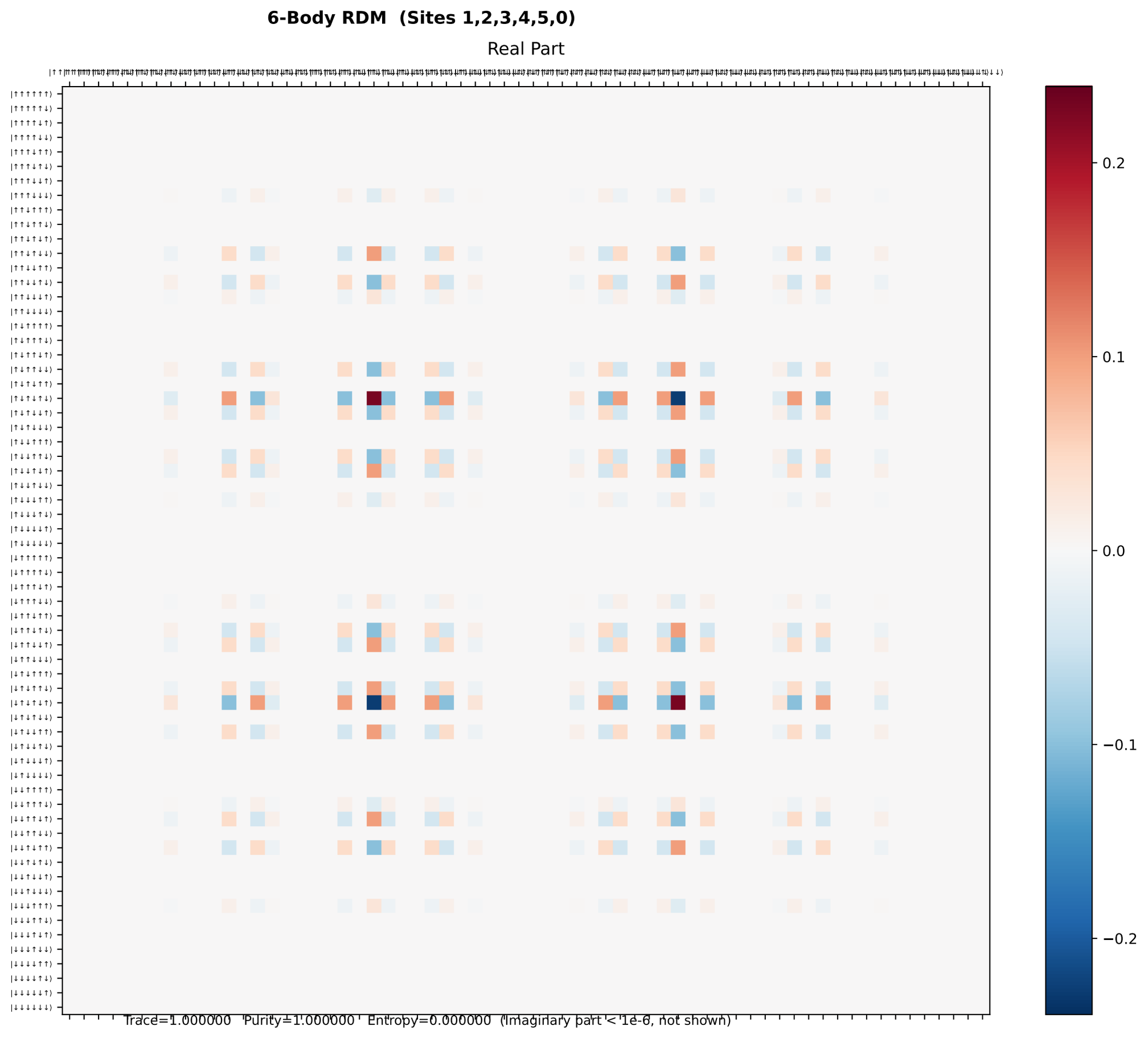
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3, 4, 5)	0.50000000
(0, 1)	(2, 3, 4, 5)	0.54089553
(0, 2)	(1, 3, 4, 5)	0.30641624
(0, 3)	(1, 2, 4, 5)	0.31883951
(0, 4)	(1, 2, 3, 5)	0.30641624
(0, 5)	(1, 2, 3, 4)	0.54089553
(0, 1, 2)	(3, 4, 5)	0.44410365
(0, 1, 3)	(2, 4, 5)	0.33307564
(0, 1, 4)	(2, 3, 5)	0.33307564
(0, 1, 5)	(2, 3, 4)	0.44410365
(0, 2, 3)	(1, 4, 5)	0.33307564
(0, 2, 4)	(1, 3, 5)	0.20962436
(0, 2, 5)	(1, 3, 4)	0.33307564
(0, 3, 4)	(1, 2, 5)	0.33307564
(0, 3, 5)	(1, 2, 4)	0.33307564
(0, 4, 5)	(1, 2, 3)	0.44410365
(0, 1, 2, 3)	(4, 5)	0.54089553
(0, 1, 2, 4)	(3, 5)	0.30641624
(0, 1, 2, 5)	(3, 4)	0.54089553
(0, 1, 3, 4)	(2, 5)	0.31883951
(0, 1, 3, 5)	(2, 4)	0.30641624
(0, 1, 4, 5)	(2, 3)	0.54089553
(0, 2, 3, 4)	(1, 5)	0.30641624
(0, 2, 3, 5)	(1, 4)	0.31883951
(0, 2, 4, 5)	(1, 3)	0.30641624
(0, 3, 4, 5)	(1, 2)	0.54089553
(0, 1, 2, 3, 4)	(5,)	0.50000000
(0, 1, 2, 3, 5)	(4,)	0.50000000
(0, 1, 2, 4, 5)	(3,)	0.50000000
(0, 1, 3, 4, 5)	(2,)	0.50000000
(0, 2, 3, 4, 5)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.


6-Body RDM (Sites 1,2,3,4,5,0)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 1,2,3,4,5,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-2.277761e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
  C_LB   = 0.45569925
C_GME (unitary optimization):
  C_stiefel (unitary opt) = 0.95823220   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.95823220

Status: nONE

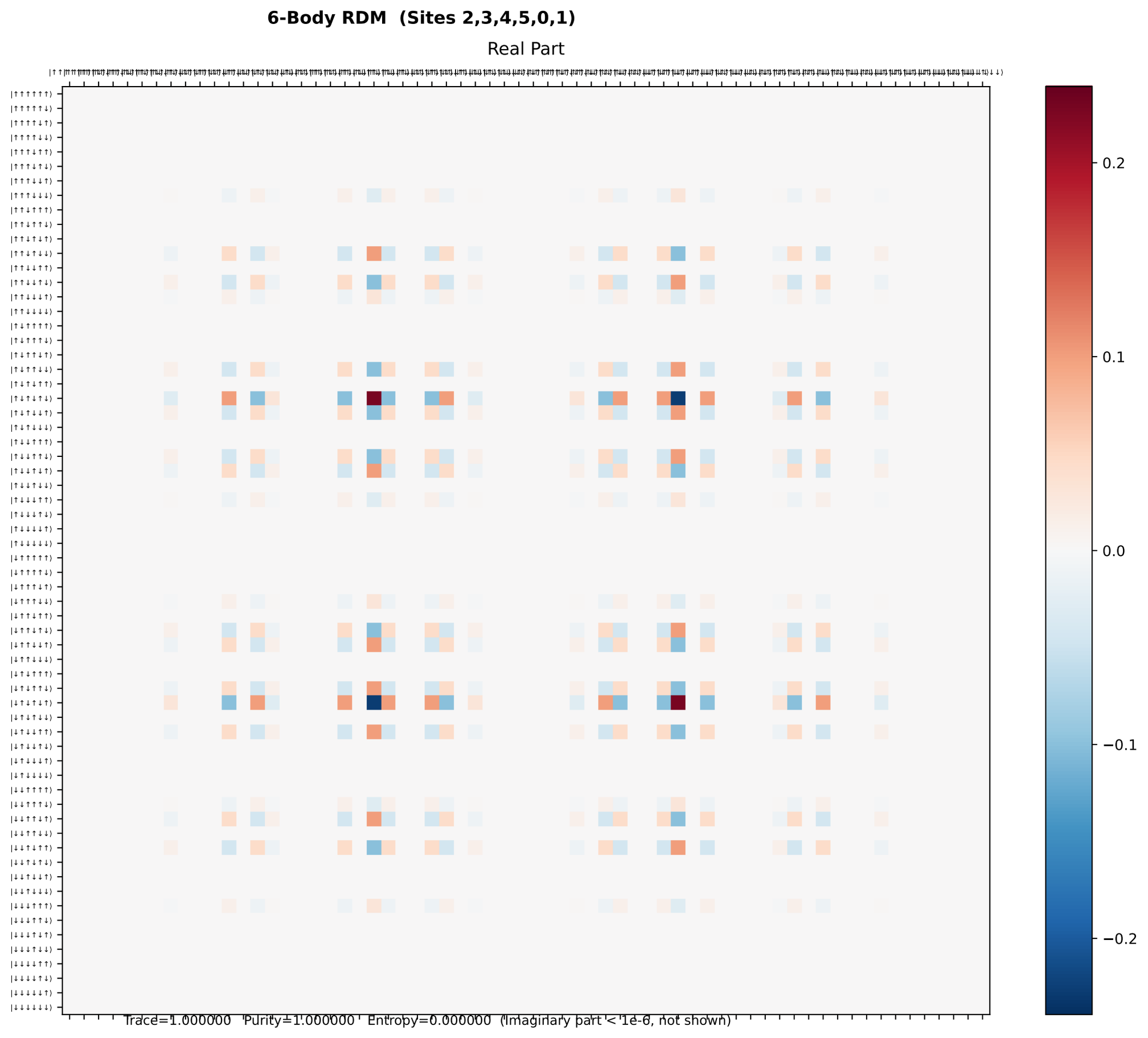
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4, 5, 0)	0.50000000
(1, 2)	(3, 4, 5, 0)	0.54089553
(1, 3)	(2, 4, 5, 0)	0.30641624
(1, 4)	(2, 3, 5, 0)	0.31883951
(1, 5)	(2, 3, 4, 0)	0.30641624
(1, 0)	(2, 3, 4, 5)	0.54089553
(1, 2, 3)	(4, 5, 0)	0.44410365
(1, 2, 4)	(3, 5, 0)	0.33307564
(1, 2, 5)	(3, 4, 0)	0.33307564
(1, 2, 0)	(3, 4, 5)	0.44410365
(1, 3, 4)	(2, 5, 0)	0.33307564
(1, 3, 5)	(2, 4, 0)	0.20962436
(1, 3, 0)	(2, 4, 5)	0.33307564
(1, 4, 5)	(2, 3, 0)	0.33307564
(1, 4, 0)	(2, 3, 5)	0.33307564
(1, 5, 0)	(2, 3, 4)	0.44410365
(1, 2, 3, 4)	(5, 0)	0.54089553
(1, 2, 3, 5)	(4, 0)	0.30641624
(1, 2, 3, 0)	(4, 5)	0.54089553
(1, 2, 4, 5)	(3, 0)	0.31883951
(1, 2, 4, 0)	(3, 5)	0.30641624
(1, 2, 5, 0)	(3, 4)	0.54089553
(1, 3, 4, 5)	(2, 0)	0.30641624
(1, 3, 4, 0)	(2, 5)	0.31883951
(1, 3, 5, 0)	(2, 4)	0.30641624
(1, 4, 5, 0)	(2, 3)	0.54089553
(1, 2, 3, 4, 5)	(0,)	0.50000000
(1, 2, 3, 4, 0)	(5,)	0.50000000
(1, 2, 3, 5, 0)	(4,)	0.50000000
(1, 2, 4, 5, 0)	(3,)	0.50000000
(1, 3, 4, 5, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.


6-Body RDM (Sites 2,3,4,5,0,1)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 2,3,4,5,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-3.101696e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.45569925
C_GME (unitary optimization):
C_stiefel (unitary opt) = 0.95823220 BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.95823220

Status: nONE

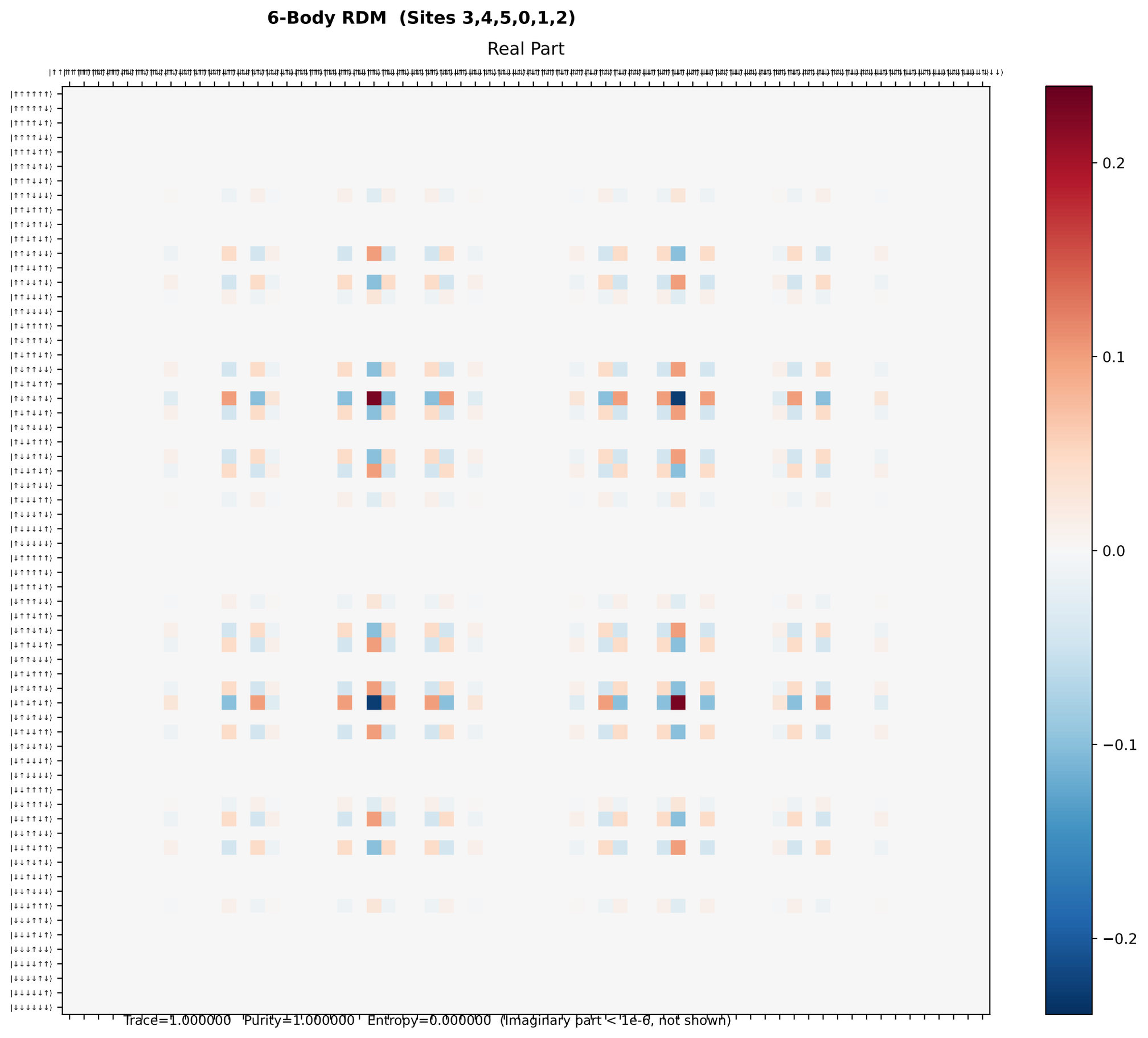
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 4, 5, 0, 1)	0.50000000
(2, 3)	(4, 5, 0, 1)	0.54089553
(2, 4)	(3, 5, 0, 1)	0.30641624
(2, 5)	(3, 4, 0, 1)	0.31883951
(2, 0)	(3, 4, 5, 1)	0.30641624
(2, 1)	(3, 4, 5, 0)	0.54089553
(2, 3, 4)	(5, 0, 1)	0.44410365
(2, 3, 5)	(4, 0, 1)	0.33307564
(2, 3, 0)	(4, 5, 1)	0.33307564
(2, 3, 1)	(4, 5, 0)	0.44410365
(2, 4, 5)	(3, 0, 1)	0.33307564
(2, 4, 0)	(3, 5, 1)	0.20962436
(2, 4, 1)	(3, 5, 0)	0.33307564
(2, 5, 0)	(3, 4, 1)	0.33307564
(2, 5, 1)	(3, 4, 0)	0.33307564
(2, 0, 1)	(3, 4, 5)	0.44410365
(2, 3, 4, 5)	(0, 1)	0.54089553
(2, 3, 4, 0)	(5, 1)	0.30641624
(2, 3, 4, 1)	(5, 0)	0.54089553
(2, 3, 5, 0)	(4, 1)	0.31883951
(2, 3, 5, 1)	(4, 0)	0.30641624
(2, 3, 0, 1)	(4, 5)	0.54089553
(2, 4, 5, 0)	(3, 1)	0.30641624
(2, 4, 5, 1)	(3, 0)	0.31883951
(2, 4, 0, 1)	(3, 5)	0.30641624
(2, 5, 0, 1)	(3, 4)	0.54089553
(2, 3, 4, 5, 0)	(1,)	0.50000000
(2, 3, 4, 5, 1)	(0,)	0.50000000
(2, 3, 4, 0, 1)	(5,)	0.50000000
(2, 3, 5, 0, 1)	(4,)	0.50000000
(2, 4, 5, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.


6-Body RDM (Sites 3,4,5,0,1,2)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 3,4,5,0,1,2)

RDM Metrics
==========

```
  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -2.199787e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

Convex Roof Analysis
==========

```
  Lower Bounds:
    C_LB    = 0.45569925
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.95823220   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.95823220

  Status: nONE
```

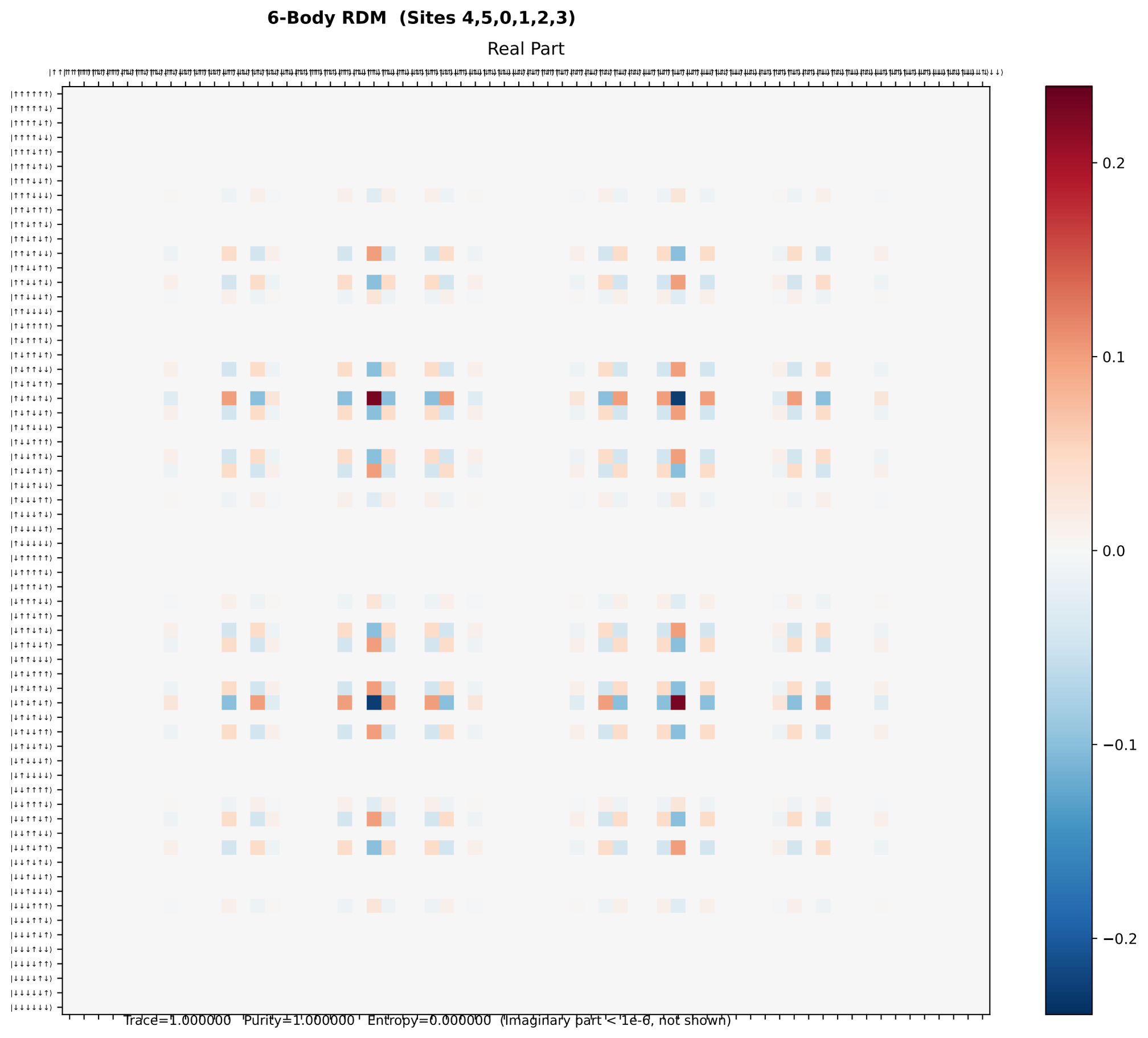
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(4, 5, 0, 1, 2)	0.50000000
(3, 4)	(5, 0, 1, 2)	0.54089553
(3, 5)	(4, 0, 1, 2)	0.30641624
(3, 0)	(4, 5, 1, 2)	0.31883951
(3, 1)	(4, 5, 0, 2)	0.30641624
(3, 2)	(4, 5, 0, 1)	0.54089553
(3, 4, 5)	(0, 1, 2)	0.44410365
(3, 4, 0)	(5, 1, 2)	0.33307564
(3, 4, 1)	(5, 0, 2)	0.33307564
(3, 4, 2)	(5, 0, 1)	0.44410365
(3, 5, 0)	(4, 1, 2)	0.33307564
(3, 5, 1)	(4, 0, 2)	0.20962436
(3, 5, 2)	(4, 0, 1)	0.33307564
(3, 0, 1)	(4, 5, 2)	0.33307564
(3, 0, 2)	(4, 5, 1)	0.33307564
(3, 1, 2)	(4, 5, 0)	0.44410365
(3, 4, 5, 0)	(1, 2)	0.54089553
(3, 4, 5, 1)	(0, 2)	0.30641624
(3, 4, 5, 2)	(0, 1)	0.54089553
(3, 4, 0, 1)	(5, 2)	0.31883951
(3, 4, 0, 2)	(5, 1)	0.30641624
(3, 4, 1, 2)	(5, 0)	0.54089553
(3, 5, 0, 1)	(4, 2)	0.30641624
(3, 5, 0, 2)	(4, 1)	0.31883951
(3, 5, 1, 2)	(4, 0)	0.30641624
(3, 0, 1, 2)	(4, 5)	0.54089553
(3, 4, 5, 0, 1)	(2,)	0.50000000
(3, 4, 5, 0, 2)	(1,)	0.50000000
(3, 4, 5, 1, 2)	(0,)	0.50000000
(3, 4, 0, 1, 2)	(5,)	0.50000000
(3, 5, 0, 1, 2)	(4,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.


6-Body RDM (Sites 4,5,0,1,2,3)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 4,5,0,1,2,3)

RDM Metrics
==========

```
  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -2.339233e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

Convex Roof Analysis
==========

```
  Lower Bounds:
    C_LB    = 0.45569925
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.95823220   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.95823220

  Status: nONE
```

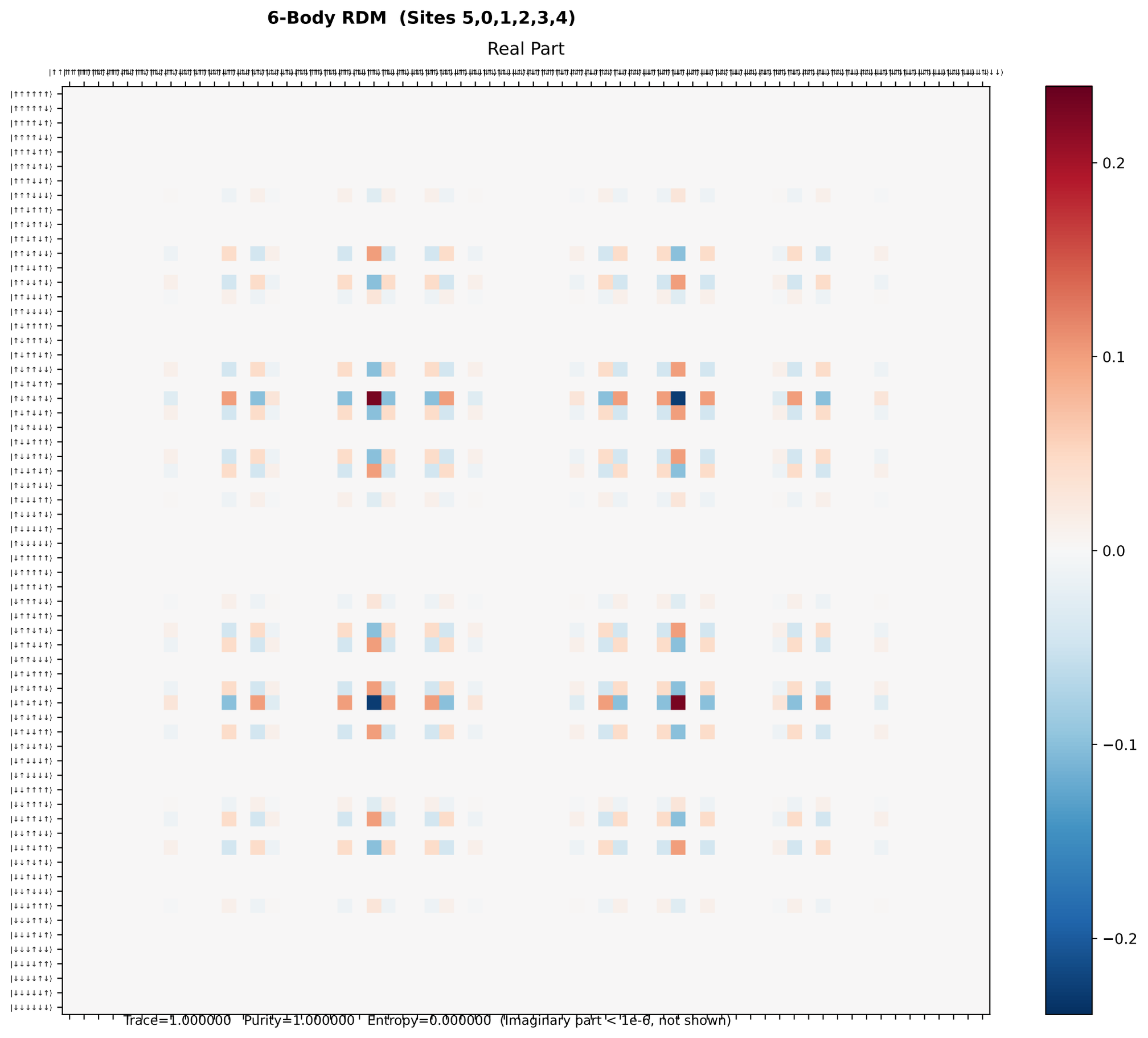
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(4,)	(5, 0, 1, 2, 3)	0.50000000
(4, 5)	(0, 1, 2, 3)	0.54089553
(4, 0)	(5, 1, 2, 3)	0.30641624
(4, 1)	(5, 0, 2, 3)	0.31883951
(4, 2)	(5, 0, 1, 3)	0.30641624
(4, 3)	(5, 0, 1, 2)	0.54089553
(4, 5, 0)	(1, 2, 3)	0.44410365
(4, 5, 1)	(0, 2, 3)	0.33307564
(4, 5, 2)	(0, 1, 3)	0.33307564
(4, 5, 3)	(0, 1, 2)	0.44410365
(4, 0, 1)	(5, 2, 3)	0.33307564
(4, 0, 2)	(5, 1, 3)	0.20962436
(4, 0, 3)	(5, 1, 2)	0.33307564
(4, 1, 2)	(5, 0, 3)	0.33307564
(4, 1, 3)	(5, 0, 2)	0.33307564
(4, 2, 3)	(5, 0, 1)	0.44410365
(4, 5, 0, 1)	(2, 3)	0.54089553
(4, 5, 0, 2)	(1, 3)	0.30641624
(4, 5, 0, 3)	(1, 2)	0.54089553
(4, 5, 1, 2)	(0, 3)	0.31883951
(4, 5, 1, 3)	(0, 2)	0.30641624
(4, 5, 2, 3)	(0, 1)	0.54089553
(4, 0, 1, 2)	(5, 3)	0.30641624
(4, 0, 1, 3)	(5, 2)	0.31883951
(4, 0, 2, 3)	(5, 1)	0.30641624
(4, 1, 2, 3)	(5, 0)	0.54089553
(4, 5, 0, 1, 2)	(3,)	0.50000000
(4, 5, 0, 1, 3)	(2,)	0.50000000
(4, 5, 0, 2, 3)	(1,)	0.50000000
(4, 5, 1, 2, 3)	(0,)	0.50000000
(4, 0, 1, 2, 3)	(5,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.


6-Body RDM (Sites 5,0,1,2,3,4)
Real Part
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 6-Body RDM (Sites 5,0,1,2,3,4)

RDM Metrics
==========

```
  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -2.024870e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

Convex Roof Analysis
==========

```
  Lower Bounds:
    C_LB    = 0.45569925
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.95823220   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.95823220

  Status: nONE
```

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(5,)	(0, 1, 2, 3, 4)	0.50000000
(5, 0)	(1, 2, 3, 4)	0.54089553
(5, 1)	(0, 2, 3, 4)	0.30641624
(5, 2)	(0, 1, 3, 4)	0.31883951
(5, 3)	(0, 1, 2, 4)	0.30641624
(5, 4)	(0, 1, 2, 3)	0.54089553
(5, 0, 1)	(2, 3, 4)	0.44410365
(5, 0, 2)	(1, 3, 4)	0.33307564
(5, 0, 3)	(1, 2, 4)	0.33307564
(5, 0, 4)	(1, 2, 3)	0.44410365
(5, 1, 2)	(0, 3, 4)	0.33307564
(5, 1, 3)	(0, 2, 4)	0.20962436
(5, 1, 4)	(0, 2, 3)	0.33307564
(5, 2, 3)	(0, 1, 4)	0.33307564
(5, 2, 4)	(0, 1, 3)	0.33307564
(5, 3, 4)	(0, 1, 2)	0.44410365
(5, 0, 1, 2)	(3, 4)	0.54089553
(5, 0, 1, 3)	(2, 4)	0.30641624
(5, 0, 1, 4)	(2, 3)	0.54089553
(5, 0, 2, 3)	(1, 4)	0.31883951
(5, 0, 2, 4)	(1, 3)	0.30641624
(5, 0, 3, 4)	(1, 2)	0.54089553
(5, 1, 2, 3)	(0, 4)	0.30641624
(5, 1, 2, 4)	(0, 3)	0.31883951
(5, 1, 3, 4)	(0, 2)	0.30641624
(5, 2, 3, 4)	(0, 1)	0.54089553
(5, 0, 1, 2, 3)	(4,)	0.50000000
(5, 0, 1, 2, 4)	(3,)	0.50000000
(5, 0, 1, 3, 4)	(2,)	0.50000000
(5, 0, 2, 3, 4)	(1,)	0.50000000
(5, 1, 2, 3, 4)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.95823220
GME confirmed across ALL cuts.